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# ASSESSMENT REPORT

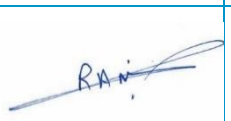
## The fire resistatance of uninsulated vertical Steel Roller Shutters

Client: SKB Shutters Manufacturing Sdn Bhd

Job number: FAS190007 Issuing consultant: Rami Al darwish

Date: 15 July 2019 Revision: R1.2

# Amendment schedule

| Version | Date | Information relating to report | | | |
|---|---|---|---|---|---|
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| | | Initial Issue | Prepared by | Reviewed by | Approved by |
| | Expiry: 30/06/2024 | Name | Rami Al Darwish | Omar Saad | Omar Saad |
| | | Signature |  |  |  |
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| | | Initial Issue | Prepared by | Reviewed by | Approved by |
| | Expiry: 30/06/2024 | Name | Rami Al Darwish | Omar Saad | Omar Saad |
| | | Signature |  |  |  |
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| | | Initial Issue | Prepared by | Reviewed by | Approved by |
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# 1. Introduction

This report presents an assessment of the fire resistance performance of uninsulated vertical roller shutters manufactured by SKB Shutters Manufacturing if tested in accordance with Singapore standard 489: 2015: Clause 5: Annex A.

The tested prototypes described in section 2 of this report, when subjected to the proposed variations described in section 3 and tested in accordance with the relevant standards described in section 4, are assessed to achieve performance as summarised in section 5.

The validity of this assessment is conditional on compliance with sections 6, 7, 8 and 9 of this report.

Summaries of the test data on which this assessment is based are provided in Appendix A. A summary of the critical issues leading to the assessment conclusions including the main points of argument is discussed in Appendix B.

# 2. Tested prototypes

This assessment is based on fire resistance test reported in 7191202724-MEC19-IHN, being a fire test conducted on an uninsulated vertical fire-resistant steel rolling shutter. The test specimen consisted of a shutter assembly of clear opening size of 2390 mm (height) × 2500 mm (width) and was mounted to a 225 mm thick brick wall construction built within the test frame. The fire resistance test was conducted in accordance with Singapore standard 489: 2015: Clause 5: Annex A. Moreover, this assessment references an additional fire resistance test, being a fire test conducted on an uninsulated vertical fire-resistant steel rolling shutter. The test specimen consisted of a shutter was mounted to a 200 mm reinforced concrete wall construction built within the test frame.

Refer to Appendix A for a full summary of the test data.

# 3. Variation to tested prototypes

## 3.1 Proposed construction specification

It is proposed that the roller shutters to be installed in openings up to 10m wide and 8m high for fire resistance periods of 60, 120 and 240 minutes. The internal construction and fixing of the roller shutter units is otherwise as tested in report 7191202724-MEC19-IHN except for the following variations:

- An increase in opening of up to 8m wide and 8m in height where possible based on available test data.
- Barrel diameter in the range of nominally as follows:
    a) 114mm × 3.3 mm
    b) 114mm × 4.5mm
    c) 114mm × 5.4mm
    d) 141mm × 4.5mm
    e) 141mm × 5.4mm
    f) 166mm × 4.5mm
    g) 166mm × 5.4mm
    h) 190mm × 6.3mm
    i) 219mm × 3.7mm
    j) 219mm × 6.3mm
    k) 241mm × 6.3mm
    l) 273mm × 4.2mm
    m) 323mm × 6.3mm
- A range of stub axle sizes. Basic specification of 35mm diameter, with consideration of larger sizes such as 40mm, 45mm, 50mm, 55mm and 60mm as appropriate.
- Various slat profile shapes, and sizes as follows:

- SKB-F38 (38mm height)
- SKB-F62 (62mm height)
- SKB-A7 (77mm height)
- SKB-A1 (92mm height)
- SKB-F7 (77mm height)
- SKB-F7D (77mm height)

- Results for fire resistance periods of 60, 120 and 240 min for galvanised and powder coated shutters
- Single or multiple barrel support frames as usually required for opening widths over 5m

**Table 1 Variation to tested prototypes**

| Item | | Size |
|---|---|---|
| **Barrel** | **Diameter (OD) (mm) × Thickness (mm)** | 114mm × 3.3mm, 4.5mm, 5.4mm |
| | | 141mm × 4.5mm, 5.4mm |
| | | 166mm × 4.5mm, 5.4mm |
| | | 190mm × 6.3mm |
| | | 219mm × 3.7mm, 6.3mm |
| | | 241mm × 6.3mm |
| | | 273mm × 4.2mm |
| | | 323mm × 6.3mm |
| | **Material** | Mild Steel (240MPa Yield Strength or high) |
| **Stub Axle** | **Diameter (mm)** | 35mm, 40mm, 45 mm, 50mm, 55mm and 60mm |
| | **Engagement in Barrel (min)** | As tested in report 7191202724-MEC19-IHN |
| | **Material** | Mild Steel (240MPa Yield Strength or high) |
| **Barrel End Plate** | **Thickness** | 4.0mm |
| | **Material** | Mild Steel (240MPa Yield Strength or high) |
| **Shutter Slat** | **Profile** | Refer to Figures 1 to 6 |
| | **Pitch** | 38 mm, 62 mm, 77 mm, and 92mm |
| | **Thickness** | 0.8mm, 1.0mm and 1.2mm |
| | **Material** | Mild Steel |
| **Bottom Bar** | **Slat Thickness** | 1.2mm |
| | **Bottom Bar** | 115mm x 2650mm length |
| | **Material** | Mild Steel |
| | **Coating for Mild Steel** | Electro-galvanized steel |
| **Barrel Support Frames** | **Up to 8m Wide** | Various refer to Figures 9-12 |
| **Guide Rail** | **Profile** | Refer to Figure 7 and 8 |
| | **Material** | Mild Steel |
| | **Fixing** | To suit expansion (see Section 7) |

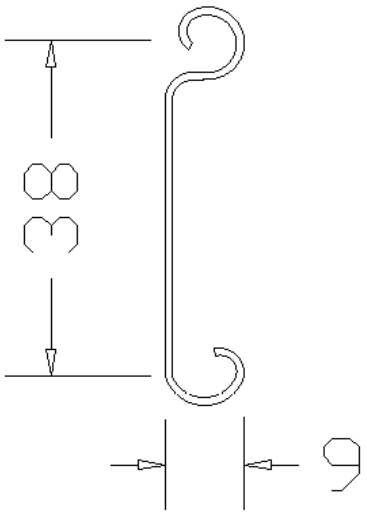



**Figure 1 Proposed Slat profile SKB-F38**

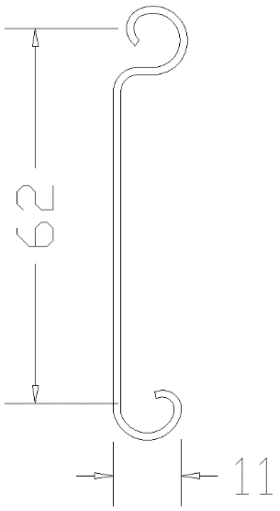



**Figure 2 Proposed Slat profile SKB-F62**




**Figure 3 Proposed Slat profile SKB-A7**

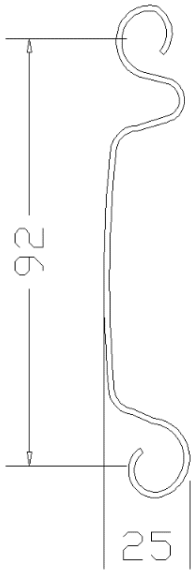



**Figure 4 Proposed Slat profile SKB-A9**

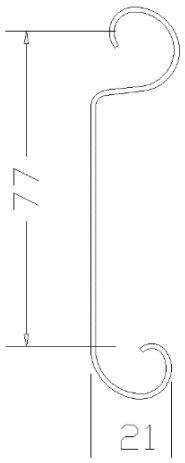



**Figure 5 Proposed Slat profile SKB-F7**

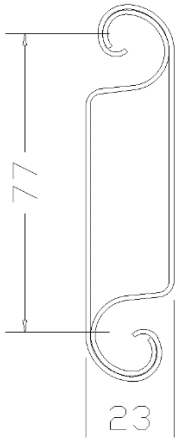



**Figure 6 Proposed Slat profile SKB-F7D**

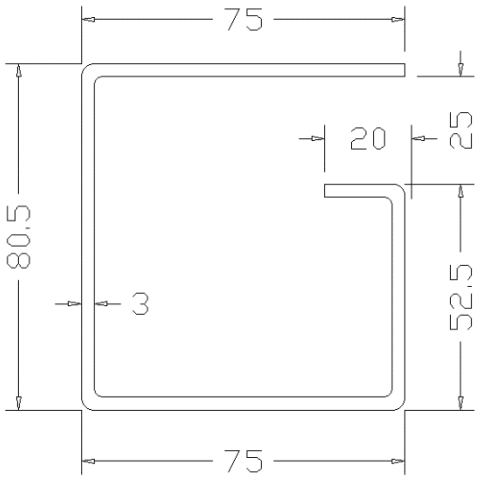



**Figure 7 Proposed Side guide SKB-G**

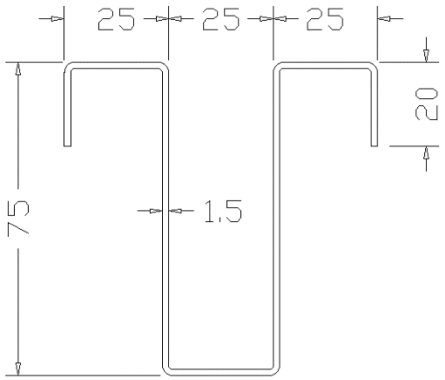
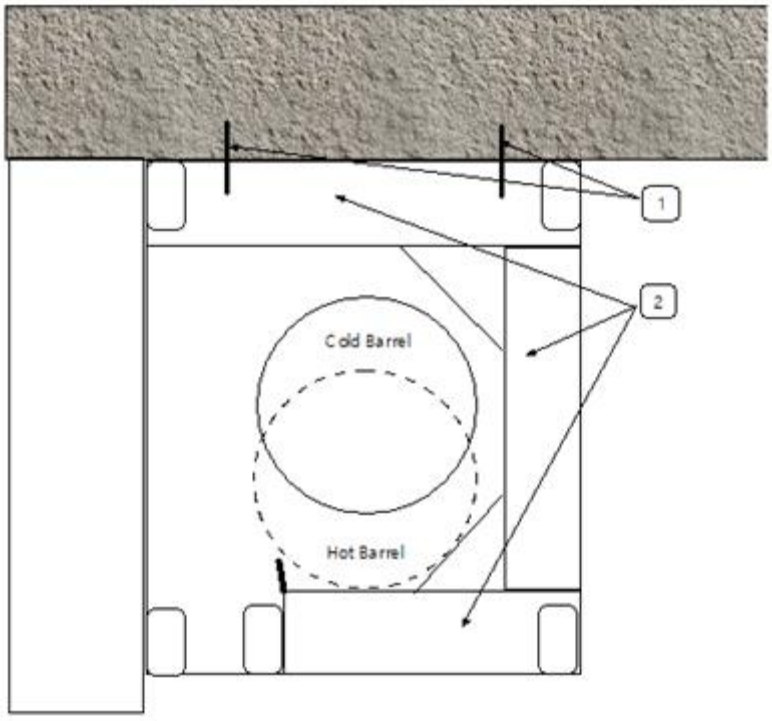
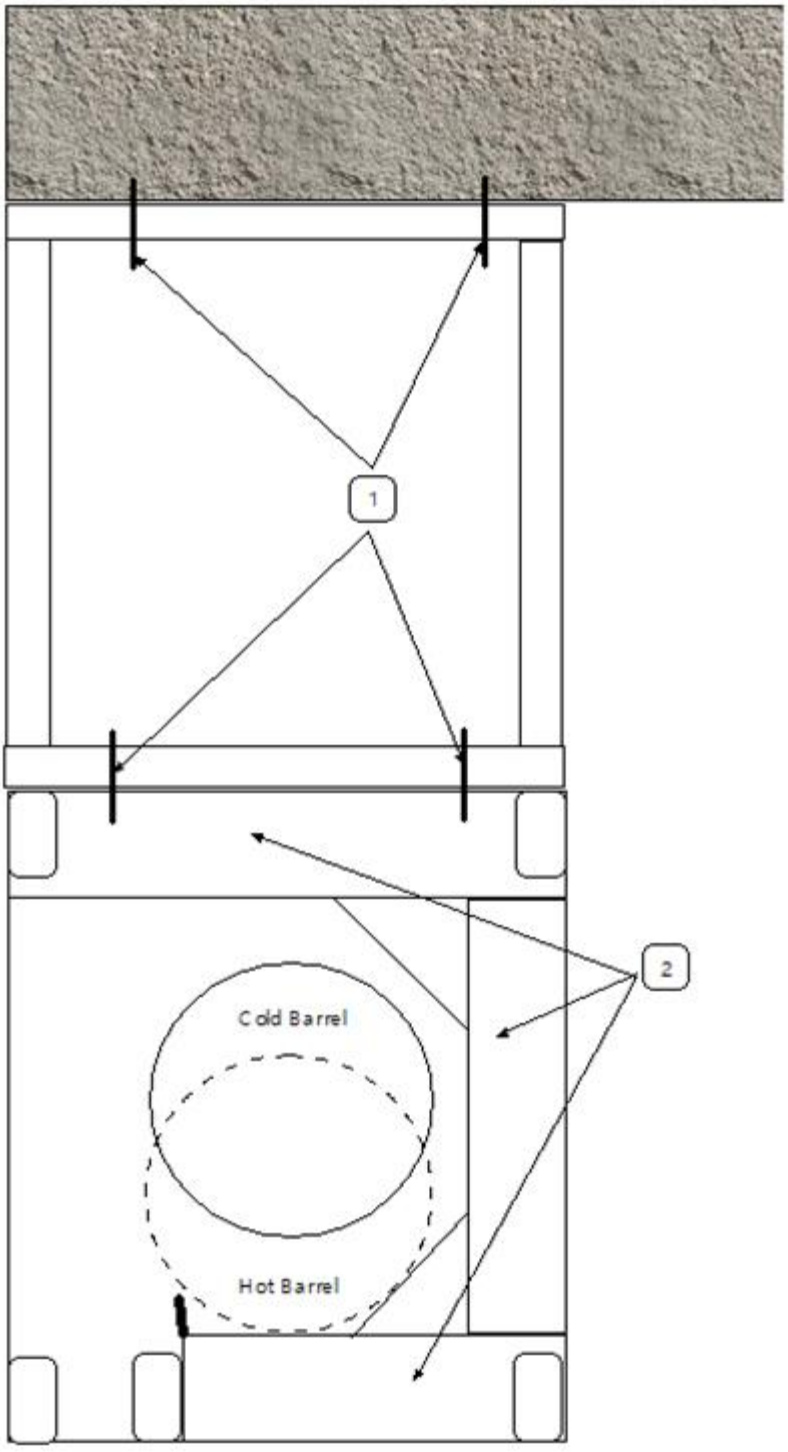



**Figure 8 Proposed Side guide SKB-U**



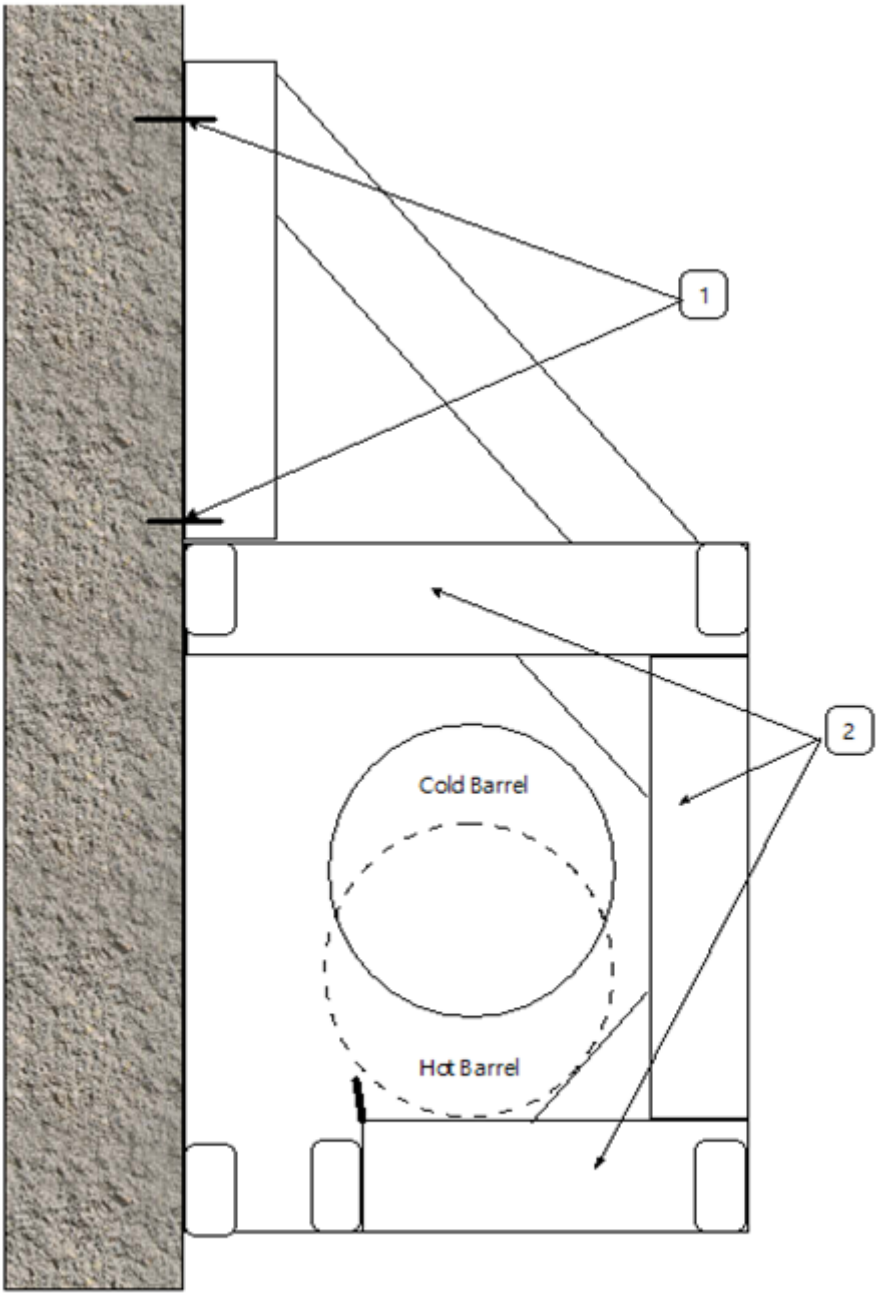

1 - End Plate Bolts
2 - 100mm x 50mm x 6mm Steel Channel Frame

**Figure 9 Proposed barrel support frame, Option 1**




1 - End Plate Bolts
2 - 100mm x 50mm x 6mm Steel Channel Frame

**Figure 10 Proposed barrel support frame, Option 2**




1 - End Plate Bolts
2 - 100mm x 50mm x 6mm Steel Channel Frame

**Figure 11 Proposed barrel support frame, Option 3**

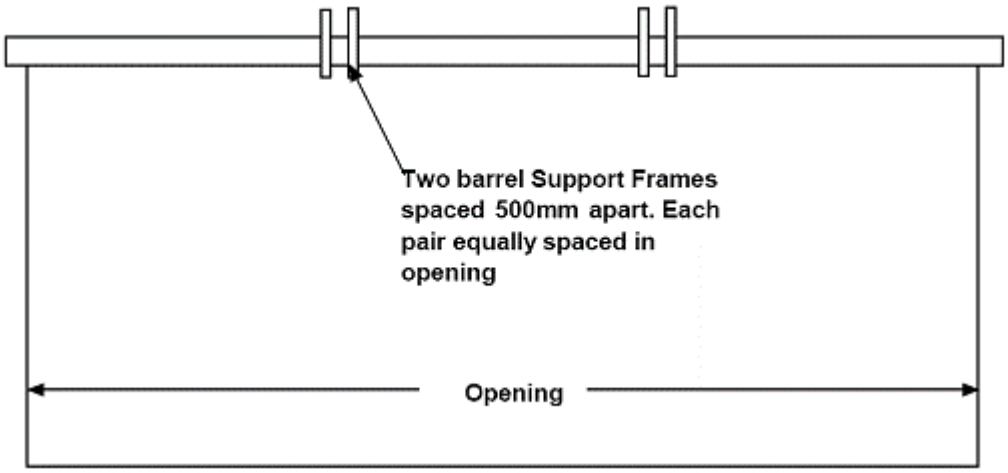



**Figure 12 Typical location of Barrel Support Frames for two pairs**

**Table 2 SKB-F38 slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-F38 | | | | | | | | FRP - 60 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 4.5 | 35 | 0.8 | 0 | M16 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 0.8 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 5.4 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | 2xM16 |
| 2.0 | 7.0 | 166 | x | 5.4 | 50 | 1.2 | 0 | 2xM16 |
| 2.0 | 8.0 | 190 | x | 6.3 | 55 | 1.2 | 0 | 2xM20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M16 |
| 4.0 | 4.0 | 114 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | 2xM16 |
| 4.0 | 8.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 2.0 | 141.0 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 5.4 | 45 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 50 | 0.8 | 1 | M20 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 8.0 | 241 | x | 6.3 | 60 | 1.2 | 1 | 3xM20 |
| 8.0 | 2.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M20 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 219.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM20 |
| 8.0 | 7.0 | 219.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |
| 8.0 | 8.0 | 219.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |

**Table 3 SKB-F38 slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-F38 | | | | | | | | FRP - 120 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 4.5 | 40 | 0.8 | 0 | M16 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 0.8 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | 2xM16 |
| 2.0 | 7.0 | 166 | x | 5.4 | 55 | 1.2 | 0 | 2xM16 |
| 2.0 | 8.0 | 190 | x | 6.3 | 55 | 1.2 | 0 | 2xM20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M16 |
| 4.0 | 4.0 | 114 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 4.0 | 5.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 55 | 1.2 | 1 | 2xM16 |
| 4.0 | 8.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 2.0 | 141.0 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 5.4 | 45 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 166.0 | x | 4.5 | 50 | 0.8 | 1 | M20 |
| 6.0 | 5.0 | 219.0 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 6.0 | 8.0 | 219 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |
| 8.0 | 2.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 0.8 | 2 | M20 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |
| 8.0 | 7.0 | 219.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |
| 8.0 | 8.0 | 219.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 4 SKB-F38 slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-F38 | | | | | | | | FRP - 240 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 4.5 | 40 | 0.8 | 0 | M16 |
| 2.0 | 4.0 | 141 | x | 5.4 | 45 | 0.8 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | 2xM16 |
| 2.0 | 7.0 | 166 | x | 5.4 | 55 | 1.2 | 0 | 2xM16 |
| 2.0 | 8.0 | 190 | x | 6.3 | 55 | 1.2 | 0 | 2xM20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M16 |
| 4.0 | 4.0 | 114 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 55 | 1.2 | 1 | 2xM16 |
| 4.0 | 8.0 | 190.0 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 2.0 | 141.0 | x | 5.4 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 5.4 | 45 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 50 | 0.8 | 1 | M20 |
| 6.0 | 5.0 | 219.0 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 241 | x | 6.3 | 60 | 1.2 | 1 | 2xM16 |
| 6.0 | 8.0 | 241 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |
| 8.0 | 2.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 0.8 | 2 | M20 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |
| 8.0 | 7.0 | 241.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 5 SKB-A7 slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-A7 | | | | | | | | FRP - 60 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M10 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 40 | 1.2 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 141 | x | 5.4 | 45 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M10 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 114 | x | 5.4 | 40 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 4.5 | 40 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 45 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | 2xM16 |
| 6.0 | 6.0 | 190 | x | 6.3 | 50 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M16 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 45 | 1.2 | 2 | M20 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 45 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 7.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |

**Table 6 SKB-A7 slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-A7 | | | | | | | | FRP - 120 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M10 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M10 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 114 | x | 5.4 | 45 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 6.0 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 4.5 | 6.0 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM16 |
| 6.0 | 6.0 | 219 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 0.8 | 2 | M16 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | M20 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 7.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 7 SKB-A7 slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-A7 | | | | | | | FRP - 240 Minutes | |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 1.2 | 0 | M12 |
| 2.0 | 3.0 | 141 | x | 4.5 | 40 | 1.2 | 0 | M16 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 4.5 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.5 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 1.2 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 5.4 | 40 | 1.2 | 1 | M16 |
| 4.0 | 4.0 | 114 | x | 5.4 | 45 | 1.2 | 1 | M16 |
| 4.0 | 4.5 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 5.5 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 166.0 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 6.0 | 3.0 | 166.0 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | 2xM16 |
| 6.0 | 4.5 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.5 | 219 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 1.2 | 2 | M16 |
| 8.0 | 4.0 | 190 | x | 6.3 | 50 | 1.2 | 2 | M20 |
| 8.0 | 4.5 | 190 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.5 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |

**Table 8 SKB-F62 slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-F62 | | | | | | | FRP - 60 Minutes | |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M10 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 141 | x | 5.4 | 45 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | 2xM16 |
| 2.0 | 8.0 | 166 | x | 5.4 | 50 | 1.2 | 0 | 2xM16 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M10 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 4.5 | 40 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 8.0 | 166.0 | x | 4.5 | 45 | 1.2 | 1 | 2xM16 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 45 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | 2xM16 |
| 6.0 | 6.0 | 219 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 8.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 45 | 0.8 | 2 | M16 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 45 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |
| 8.0 | 7.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |
| 8.0 | 8.0 | 241.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |

**Table 9 SKB-F62 slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-F62 | | | | | | | FRP - 120 Minutes | |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M10 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | 2xM16 |
| 2.0 | 8.0 | 166 | x | 5.4 | 55 | 1.2 | 0 | 2xM16 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M10 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 8.0 | 166.0 | x | 4.5 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 6.0 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 4.5 | 6.0 | 0.8 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM16 |
| 6.0 | 6.0 | 219 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM20 |
| 6.0 | 8.0 | 219 | x | 6.3 | 6.0 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 8.0 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 8.0 | 0.8 | 2 | M16 |
| 8.0 | 4.0 | 241.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 5.4 | 8.0 | 1.2 | 2 | 2xM16 |
| 8.0 | 7.0 | 241.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM20 |
| 8.0 | 8.0 | 219.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM20 |

**Table 10 SKB-F62 slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-F62 | | | | | | | | FRP - 240 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 1.2 | 0 | M10 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 1.2 | 0 | M12 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.5 | 166 | x | 4.5 | 50 | 1.2 | 0 | 2xM16 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 1.2 | 1 | M10 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 1.2 | 1 | M12 |
| 4.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 166 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 6.5 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 1.2 | 1 | M16 |
| 6.0 | 3.0 | 141.0 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.5 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 1.2 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 1.2 | 2 | M16 |
| 8.0 | 4.0 | 241.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |
| 8.0 | 6.5 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 11 SKB-A9 slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-A9 | | | | | | | | FRP - 60 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M16 |
| 2.0 | 5.0 | 141 | x | 5.4 | 45 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M16 |
| 4.0 | 4.0 | 141 | x | 4.5 | 40 | 1.2 | 1 | M16 |
| 4.0 | 5.0 | 141 | x | 4.5 | 40 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 190.0 | x | 6.3 | 45 | 1.2 | 1 | M20 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 45 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 50 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 190 | x | 6.3 | 8.0 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 219 | x | 6.3 | 8.0 | 1.2 | 2 | M20 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 8.0 | 1.2 | 2 | M20 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 8.0 | 1.2 | 2 | 2xM20 |

**Table 12 SKB-A9 slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-A9 | | | | | | | | FRP - 120 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 1.2 | 0 | M12 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.0 | 166 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 1.2 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 166.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 166.0 | x | 4.5 | 40 | 1.2 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | M16 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 1.2 | 2 | M16 |
| 8.0 | 4.0 | 241.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 241.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 241.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |

**Table 13 SKB-A9 slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-A9 | | | | | | | | FRP - 240 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M12 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 4.25 | 166 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 141 | x | 4.5 | 40 | 1.2 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 4.25 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 166.0 | x | 5.4 | 45 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 190.0 | x | 6.3 | 50 | 1.2 | 1 | M16 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 4.25 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 219 | x | 6.3 | 55 | 1.2 | 2 | M16 |
| 8.0 | 4.0 | 241.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 4.25 | 241.0 | x | 6.3 | 50 | 1.2 | 2 | 2xM16 |

**Table 14 SKB-F7 slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-F7 | | | | | | | | FRP - 60 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 5.4 | 45 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 5.4 | 45 | 1.2 | 0 | 2xM16 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 4.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 7.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | 2xM16 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 190.0 | x | 6.3 | 45 | 0.8 | 1 | M20 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 0.8 | 2 | M20 |
| 8.0 | 4.0 | 190 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM20 |
| 8.0 | 7.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 15 SKB-F7 slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-F7 | | | | | | | | FRP - 120 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 0 | M20 |
| 2.0 | 7.0 | 166 | x | 5.4 | 55 | 1.2 | 0 | 2xM16 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 3.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 4.0 | 5.0 | 141 | x | 4.5 | 45 | 1.2 | 1 | M20 |
| 4.0 | 6.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 5.0 | 166 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 141.0 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 6.0 | 3.0 | 190.0 | x | 6.3 | 45 | 0.8 | 1 | M20 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 6.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 6.0 | 7.0 | 219 | x | 6.3 | 60 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 3.0 | 190 | x | 6.3 | 50 | 0.8 | 2 | M20 |
| 8.0 | 4.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | M20 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 6.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |
| 8.0 | 7.0 | 190.0 | x | 6.3 | 60 | 1.2 | 2 | 2xM20 |

**Table 16 SKB-F7 slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-F7 | | | | | | | FRP - 240 Minutes | |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 2.5 | 114 | x | 3.3 | 35 | 0.8 | 0 | M12 |
| 2.0 | 4.0 | 141 | x | 4.5 | 45 | 1.2 | 0 | M20 |
| 2.0 | 5.0 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 2.0 | 5.5 | 141 | x | 5.4 | 50 | 1.2 | 0 | M20 |
| 4.0 | 2.0 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 2.5 | 114 | x | 3.3 | 35 | 0.8 | 1 | M12 |
| 4.0 | 4.0 | 141 | x | 5.4 | 45 | 1.2 | 1 | M20 |
| 4.0 | 5.0 | 141 | x | 4.5 | 50 | 1.2 | 1 | M20 |
| 4.0 | 5.5 | 141 | x | 5.4 | 50 | 1.2 | 1 | M20 |
| 6.0 | 2.0 | 139.8 | x | 4.0 | 40 | 0.8 | 1 | M16 |
| 6.0 | 2.5 | 190.0 | x | 6.3 | 45 | 0.8 | 1 | M20 |
| 6.0 | 4.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM16 |
| 6.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 6.0 | 5.5 | 190 | x | 6.3 | 55 | 1.2 | 1 | 2xM20 |
| 8.0 | 2.0 | 166 | x | 5.4 | 45 | 0.8 | 2 | M16 |
| 8.0 | 2.5 | 190 | x | 6.3 | 50 | 0.8 | 2 | M16 |
| 8.0 | 4.0 | 190 | x | 6.3 | 60 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.0 | 190.0 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |
| 8.0 | 5.5 | 190 | x | 6.3 | 55 | 1.2 | 2 | 2xM16 |

**Table 17 SKB-F7D slat roller shutter configurations for 60 minutes of FRP**

| Profile: SKB-F7D | | | | | | | | FRP - 60 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 166 | x | 4.5 | 40 | 0.8 | 0 | M16 |
| 2.0 | 3.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 5.0 | 166 | x | 4.5 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 7.0 | 190 | x | 6.3 | 55 | 0.8 | 0 | 2XM20 |
| 2.0 | 8.0 | 190 | x | 6.3 | 55 | 0.8 | 0 | 2XM20 |
| 4.0 | 2.0 | 166 | x | 4.5 | 35 | 0.8 | 1 | M16 |
| 4.0 | 3.0 | 166 | x | 4.5 | 40 | 0.8 | 1 | M20 |
| 4.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | M20 |
| 4.0 | 5.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | 2xM16 |
| 4.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 1 | 2xM16 |
| 4.0 | 7.0 | 190 | x | 6.3 | 50 | 0.8 | 1 | 2xM20 |
| 4.0 | 8.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2xM20 |
| 6.0 | 2.0 | 190 | x | 6.3 | 45 | 0.8 | 1 | M20 |
| 6.0 | 3.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2xM16 |
| 6.0 | 4.0 | 219 | x | 6.3 | 50 | 0.8 | 1 | 2xM20 |
| 6.0 | 5.0 | 219 | x | 6.3 | 55 | 0.8 | 1 | 2xM16 |
| 6.0 | 6.0 | 241 | x | 6.3 | 55 | 0.8 | 1 | 2XM20 |
| 6.0 | 7.0 | 241 | x | 6.3 | 60 | 0.8 | 1 | 2XM25 |
| 6.0 | 8.0 | 323 | x | 6.3 | 60 | 0.8 | 1 | 2XM25 |
| 8.0 | 2.0 | 219 | x | 6.3 | 50 | 0.8 | 2 | M20 |
| 8.0 | 3.0 | 241 | x | 6.3 | 55 | 0.8 | 2 | 2xM16 |
| 8.0 | 4.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 5.0 | 323 | x | 6.3 | 55 | 0.8 | 2 | 2xM20 |
| 8.0 | 6.0 | 323 | x | 6.3 | 55 | 0.8 | 2 | 2XM20 |
| 8.0 | 7.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2XM25 |
| 8.0 | 8.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2XM25 |

**Table 18 SKB-F7D slat roller shutter configurations for 120 minutes of FRP**

| Profile: SKB-F7D | | | | | | | FRP - 120 Minutes | |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 166 | x | 4.5 | 40 | 0.8 | 0 | M16 |
| 2.0 | 3.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 5.0 | 166 | x | 4.5 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 7.0 | 190 | x | 6.3 | 55 | 0.8 | 0 | 2xM20 |
| 2.0 | 8.0 | 190 | x | 6.3 | 55 | 0.8 | 0 | 2xM20 |
| 4.0 | 2.0 | 166 | x | 4.5 | 35 | 0.8 | 1 | M16 |
| 4.0 | 3.0 | 166 | x | 4.5 | 40 | 0.8 | 1 | M20 |
| 4.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | M20 |
| 4.0 | 5.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | 2xM16 |
| 4.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 1 | 2xM16 |
| 4.0 | 7.0 | 190 | x | 6.3 | 50 | 0.8 | 1 | 2XM20 |
| 4.0 | 8.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2XM20 |
| 6.0 | 2.0 | 190 | x | 6.3 | 45 | 0.8 | 1 | M20 |
| 6.0 | 3.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2xM16 |
| 6.0 | 4.0 | 219 | x | 6.3 | 50 | 0.8 | 1 | 2xM20 |
| 6.0 | 5.0 | 219 | x | 6.3 | 55 | 0.8 | 1 | 2xM20 |
| 6.0 | 6.0 | 323 | x | 6.3 | 65 | 0.8 | 1 | 2xM20 |
| 6.0 | 7.0 | 241 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 6.0 | 8.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 2.0 | 219 | x | 6.3 | 55 | 0.8 | 2 | M20 |
| 8.0 | 3.0 | 241 | x | 6.3 | 60 | 0.8 | 2 | 2xM16 |
| 8.0 | 4.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 5.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 6.0 | 219 | x | 6.3 | 60 | 0.8 | 3 | 2xM20 |
| 8.0 | 7.0 | 219 | x | 6.3 | 60 | 0.8 | 3 | 2xM20 |
| 8.0 | 8.0 | 241 | x | 6.3 | 60 | 0.8 | 3 | 2xM20 |

**Table 19 SKB-F7D slat roller shutter configurations for 240 minutes of FRP**

| Profile: SKB-F7D | | | | | | | | FRP - 240 Minutes |
|---|---|---|---|---|---|---|---|---|
| Opening Width (m) | Opening Height (m) | Barrel Size (ID) x Wall (mm) | | | Shaft OD (mm) | Slat thickness | Paired Barrel Support Frames | End Plate Bolts |
| 2.0 | 2.0 | 166 | x | 4.5 | 40 | 0.8 | 0 | M16 |
| 2.0 | 3.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 0 | M20 |
| 2.0 | 5.0 | 166 | x | 4.5 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 0 | 2xM16 |
| 2.0 | 7.0 | 190 | x | 6.3 | 60 | 0.8 | 0 | 2xM20 |
| 4.0 | 2.0 | 166 | x | 4.5 | 40 | 0.8 | 1 | M16 |
| 4.0 | 3.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | M20 |
| 4.0 | 4.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | M20 |
| 4.0 | 5.0 | 166 | x | 4.5 | 45 | 0.8 | 1 | 2xM16 |
| 4.0 | 6.0 | 166 | x | 5.4 | 50 | 0.8 | 1 | 2xM16 |
| 4.0 | 7.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2XM20 |
| 4.0 | 8.0 | 190 | x | 6.3 | 60 | 0.8 | 1 | 2XM20 |
| 6.0 | 2.0 | 190 | x | 6.3 | 50 | 0.8 | 1 | M20 |
| 6.0 | 3.0 | 190 | x | 6.3 | 55 | 0.8 | 1 | 2xM16 |
| 6.0 | 4.0 | 219 | x | 6.3 | 55 | 0.8 | 1 | 2xM20 |
| 6.0 | 5.0 | 219 | x | 6.3 | 55 | 0.8 | 1 | 2xM20 |
| 6.0 | 6.0 | 190 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 6.0 | 7.0 | 190 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 2.0 | 219 | x | 6.3 | 55 | 0.8 | 2 | M20 |
| 8.0 | 3.0 | 241 | x | 6.3 | 60 | 0.8 | 2 | 2xM16 |
| 8.0 | 4.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 5.0 | 323 | x | 6.3 | 60 | 0.8 | 2 | 2xM20 |
| 8.0 | 6.0 | 219 | x | 6.3 | 60 | 0.8 | 3 | 2xM20 |
| 8.0 | 7.0 | 241 | x | 6.3 | 60 | 0.8 | 3 | 2xM20 |

# 4. Referenced test standard

This report is prepared with reference to the requirements of Singapore standard 489: 2015: Clause 5: Annex A

# 5. Formal assessment summary

On the basis of the discussion presented in this report, it is the opinion of this testing authority that if the tested prototype described in section 2 had been varied as in section 3, it would achieve the integrity fire resistance periods of 60, 120 and 240 minutes if tested in accordance with the test method referenced in section 4 when subject to the requirements of section 7.

# 6. Direct field of application

This assessment applies to roller shutter assemblies smaller than the assessed size provided there are no other changes to the construction or installation details of the components. Further, Barrel size and shaft diameters stipulated in Tables 2-19 are the minimum sizes to achieve the corresponding fire resistance period. Hence, larger barrel sizes or shaft diameters than the stipulated can be used.

This assessment is based on details recorded during a fire resistance test where the roller shutter assembly and barrel was exposed to the side considered likely to provide the least period of fire resistance. The assessment is therefore valid for fire exposure from either side.

# 7. Requirements

This report details the methods of construction, test conditions and assessed results that would have been expected had the specific elements of construction described herein been tested in accordance with Singapore standard 489: 2015: Clause 5: Annex A

The supporting masonry or concrete construction shall be capable of providing effective support for the proposed construction for the required fire period.

It is required that the contribution to the curtain support by the floor be consistent throughout the range of door openings considered.

The required minimum yield stress for components made from steel or stainless steel with a minimum yield strength of 240MPa.

The calculations referred to this report concern the expected stability of the critical components at elevated temperatures when the shutter is in the closed position. The recommended sizes of components may not necessarily be appropriate for shutter under normal operating temperatures where safety and other loadings may need to be considered.

The calculations referred to this report concern the expected expansion of the critical components at elevated temperatures that are presented below. The installation and adjustment of the proposed construction shall allow for following expansion of each component.

**Expected expansion at each end (mm) up to 60mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 14 | 27 | 41 | 54 |
| Curtain | 12 | 24 | 36 | 48 |
| Bottom Rail | 4 | 7 | 11 | 14 |

**Expected expansion at each end (mm) up to 120mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 16 | 31 | 46 | 61 |
| Curtain | 1 | 24 | 36 | 48 |
| Bottom Rail | 6 | 12 | 18 | 24 |

**Expected expansion at each end (mm) up to 240mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 16 | 32 | 48 | 64 |
| Curtain | 18 | 36 | 54 | 72 |
| Bottom Rail | 10 | 21 | 31 | 41 |

**Expected guide rail expansion at end (mm) for 60, 120 and 240mins FRP**

| Component | Expansion (mm) for Given Door height | | | | | | |
|---|---|---|---|---|---|---|---|
| | 2000 | 3000 | 4000 | 5000 | 6000 | 7000 | 8000 |
| 60 Minutes | 11 | 16 | 21 | 26 | 31 | 36 | 40 |
| 120 Minutes | 13 | 18 | 24 | 29 | 35 | 41 | 46 |
| 240 Minutes | 13 | 19 | 22 | 30 | 36 | 42 | 48 |

The minimum engagement of the curtain in the guide rail shall be maintained as tested in the referenced tests for all design options listed in Section 3.

Any further variations with respect to size, constructional details, loads, stresses, edge or end conditions, other than those identified in this report, may invalidate the conclusions drawn in this report.

# 8. Validity

This assessment report does not provide an endorsement by Warringtonfire Aus Pty Ltd of the actual products supplied.

The conclusions of this assessment may be used to directly assess fire hazard, but it should be recognised that a single test method will not provide a full assessment of fire hazard under all conditions.

Because of the nature of fire testing, and the consequent difficulty in quantifying the uncertainty of measurement, it is not possible to provide a stated degree of accuracy. The inherent variability in test procedures, materials and methods of construction, and installation may lead to variations in performance between elements of similar construction.

The assessment can therefore only relate only to the actual prototype test specimens, testing conditions, and methodology described in the supporting data, and does not imply any performance abilities of constructions of subsequent manufacture.

This assessment is based on information and experience available at the time of preparation. The published procedures for the conduct of tests and the assessment of test results are the subject of constant review and improvement and it is recommended that this report be reviewed on or, before, the stated expiry date.

The information contained in this report shall not be used for the assessment of variations other than those stated in the conclusions above. The assessment is valid provided no modifications are made to the systems detailed in this report. All details of construction should be consistent with the requirements stated in the relevant test reports and all referenced documents.

# 9. Authority

## 9.1 Applicant undertakings and conditions of use

By using this report as evidence of compliance or performance, the applicant(s) confirms that:

- To their knowledge the component or element of structure, which is the subject of this assessment, has not been subjected to a fire test to the standard against which this assessment is being made, and

- They agree to withdraw this assessment from circulation should the component or element of structure be the subject of a fire test by a test authority in accordance with the standard against which this assessment is being made and the results are not in agreement with this assessment, and
- They are not aware of any information that could adversely affect the conclusions of this assessment and if they subsequently become aware of any such information, agree to ask the assessing authority to withdraw the assessment.

## 9.2 General conditions of use

This report may only be reproduced in full without modifications by the report sponsor. Copies, extracts or abridgments of this report in any form shall not be published by other organisations or individuals without the permission of Warringtonfire Aus Pty Ltd.

# Appendix A Summary of supporting data

## A.1 Test report – 7191202724-MEC19-IHN

### A.1.1 Report sponsor

SKB Shutters Manufacturing Sdn Bhd, Lot 22, Jalan Teknologi, Taman Sains Selangor1, Kota Damasara, 47810 Petaling Jaya, Selangor Darul Ehsan, Malaysia

### Test laboratory

TUV SUD PSB Pte.Ltd. No.1 Science Park Drive, Singapore 118221

### A.1.2 Test date

The fire resistance test was conducted on 14/01/2019.

### A.1.3 Test standards

The test was conducted in accordance with Singapore standard 489: 2015: Clause 5: Annex A

### A.1.4 Variations to test standards

None

### A.1.5 General description of tested specimen

The test specimen consisted of an uninsulated SKB Roller Shutter assembly installed over a brick wall with a clear opening of 2500mm wide × 2350mm high. Overall size of the shutter assembly was 2670mm wide × 2740mm high. The side guides and shutter box were surface mounted and mechanically anchored to surrounding wall with M10 and M12 anchor bolts. The wall surrounding the test specimen was constructed using ordinary bricks with a concrete lintel and the assembly was installed with all its relevant components exposed to the fire. It was constructed within a test frame and mounted onto the test furnace.

### A.1.6 Test results

The ambient temperature at the beginning of the test and on completion of the test was between 10°C and 40°C.

The tested prototype met integrity performance requirements of the standard for a period of 260 minutes without collapse, sustained flaming or lack of impermeability.

The temperatures of structurally critical components are listed below.

| Location | Thermocouple | 0 | 60 | 120 | 180 | 240 |
|---|---|---|---|---|---|---|
| **Shaft** | T-16, T-18 | 28.4 | 933.7 | 1060 | 1067.9 | 1099.2 |
| **Endplate** | T-17 , T-19 | 28.7 | 937.3 | 1056.1 | 1080.3 | 1105.4 |
| **Barrel** | T-15 | 28.4 | 978.8 | 1061.1 | 1077.6 | 1108.5 |
| **Curtain** | TC-6,TC-7,TC-8,TC-9,TC-10 | 28.4 | 906.1 | 906.1 | 1252.6 | 1200 |
| **Guide Rail** | T-13, T-14 | 27.8 | 674.3 | 805.3 | 850.3 | 878.5 |
| **Bottom Rail** | T11 | 27.2 | 288.5 | 447.4 | 608.6 | 710.2 |

## A.2 Additional Supporting Test Evidence

### A.2.1 Report sponsor

SKB Shutters Manufacturing Sdn Bhd, Lot 22, Jalan Teknologi, Taman Sains Selangor1, Kota Damasara, 47810 Petaling Jaya, Selangor Darul Ehsan, Malaysia

### A.2.2 Test date

The fire resistance test was conducted on 28/06/2018.

### A.2.3 Variations to test standards

None

### A.2.4 General description of tested specimen

An uninsulated vertical fire resistant steel roller shutter assembly was submitted to the Fire Protection Section on 11 February 2019. The test specimen consisted of a motor driven steel roller shutter assembly which was mounted into a 200 mm thick reinforced concrete construction built within a test frame lined with refractory concrete. The shutter curtain was mounted on the face of the wall within the guide rails and operated freely before the test. The mild steel barrel built on the face of the wall which was to face the heating condition of the test. The guide rails assembly together with the frame structure (barrel casing box) was secured to the wall with expanding anchor bolts. A precast concrete floor slab was provided as part of the opening to be protected and projected into the furnace approximately 100 mm. Any gaps and voids between the specimen and concrete were finished with mortar. Inspection was carried out during the construction of the specimen to verify on its design, dimensions and materials used.

### A.2.5 Test results

Integrity: 250 minutes

The test was terminated after a period of 250 minutes at the request of the applicant.

# Appendix B Assessment of specific variations

## B.1 Expected expansion of components

### B.1.1 Expansion of roller shutter components

1. Due to the increase in the width and height of the shutter curtain, an increase in the thermal expansion of all components is expected. With reference to the temperature of each components at various fire resistance periods and the expansion coefficients in BS 5950 Part 8, the expected expansion at each end was calculated and are presented below:

**Table B1 Expected expansion at each end (mm) up to 60mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 14 | 27 | 41 | 54 |
| Curtain | 12 | 24 | 36 | 48 |
| Bottom Rail | 4 | 7 | 11 | 14 |

**Table B2 Expected expansion at each end (mm) up to 120mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 16 | 31 | 46 | 61 |
| Curtain | 1 | 24 | 36 | 48 |
| Bottom Rail | 6 | 12 | 18 | 24 |

**Table B3 Expected expansion at each end (mm) up to 240mins FRP**

| Component | Expansion (mm) for Given Door Width | | | |
|---|---|---|---|---|
| | 2000 | 4000 | 6000 | 8000 |
| Barrel/Axle | 16 | 32 | 48 | 64 |
| Curtain | 18 | 36 | 54 | 72 |
| Bottom Rail | 10 | 21 | 31 | 41 |

**Table B4 Expected guide rail expansion at end (mm) for 60, 120 and 240mins FRP**

| Component | Expansion (mm) for Given Door height | | | | | | |
|---|---|---|---|---|---|---|---|
| | 2000 | 3000 | 4000 | 5000 | 6000 | 7000 | 8000 |
| 60 Minutes | 11 | 16 | 21 | 26 | 31 | 36 | 40 |
| 120 Minutes | 13 | 18 | 24 | 29 | 35 | 41 | 46 |
| 240 Minutes | 13 | 19 | 22 | 30 | 36 | 42 | 48 |

2. During the installation of the shutter, the calculated expansions values shown above must be accommodated. If inadequate allowance is made for thermal expansion, the result may be buckling of some components and possible premature failure of the shutter.

3. Based on the above discussion it is considered that the proposed design will not likely introduce any integrity or impermeability weakness to the proposed shutter configurations for up to 240 minutes, provided the above expansion allowances are made during the installation of the proposed shutter.

## B.2 Variation to shutter curtain design

### B.2.1 General

4. The important aspect of the fire resistance performance of the proposed shutter curtain slats is based on the ability to remain in place and not split or deform in any way that may create an opening leading to an integrity failure with respect to the referenced test standards.

5. For the proposed construction, failure is deemed to have occurred, if during the test period:

- A 6mm × 150 gap gauge is able penetrate a through gap into the furnace that has formed during the test
- A 25mm diameter gap gauge is able penetrate a through gap into the furnace that has formed during the test
- The specimen collapses during the test period
- Sustained flaming for longer than 10s is observed.

**Table B5 Proposed maximum shutter curtain heights**

| Slat Name | Shutter Slat Thickness (mm) | Curtain Height | | |
|---|---|---|---|---|
| | | 60 minutes | 120 minutes | 240 minutes |
| SKB-F38 | 0.8mm to 1.2mm | 8000 | 8000 | 8000 |
| SKB-F62 | 0.8mm to 1.2mm | 8000 | 8000 | 6500 |
| SKB-A7 | 0.8mm to 1.2mm | 7000 | 7000 | 5500 |
| SKB-A9 | 0.8mm to 1.2mm | 6000 | 6000 | 4250 |
| SKB-F7 | 0.8mm to 1.2mm | 7000 | 7000 | 5500 |
| SKB-F7D | 0.8mm to 1.2mm | 8000 | 8000 | 7000 |

### B.2.2 Discussion

**Shutter Curtain Slat Profile**

6. The proposed curtain profile in Figure 1 to 6 have similar scroll profiles at the point at which they interlock. This feature ensures that when hanging vertically, the stress levels in the scroll section will also be very similar and only vary in proportion to the size of the scroll.

7. The difference in slat thickness does not introduce any other strength or integrity related weakness given that the maximum allowable stress criterion is met.

8. In absence of any strength related weakness, it is considered the proposed slat profiles would perform similarly if tested in accordance with the reference standard and therefore both meet the proposed fire-resistant periods.

**Shutter Curtain Height**

9. With reference to the tested construction summarised in Appendix A, the tested shutter curtain satisfactorily supported its own weight in excess of 240 minutes.

10. The proposed construction incorporates extension of height of the roller shutter for periods of exposure of 60, 120 and 240 minutes the heights shown in Table B5.

11. Confidence in this proposal for periods less than 240 minutes is given by the fact that the shutter curtain is cooler at these times and therefore the shutter will retain more of its yield strength.

12. Confidence in this proposal for periods of up to 240 minutes is given by the fact that thicker, stronger profiles are used for higher shutter curtains.

13. With reference to BS 5950 Part 8 and the results of the reference tests, the strength retention at 60, 120 and 240 minutes were calculated, and the height increased until the stress ratio was that of the tested construction or the residual strength predicted by BS 5950 Part 8.

14. In absence of any introduced strength related weakness of the proposed construction, it is considered the proposed increased heights for the proposed construction will meet the relevant integrity and impermeability requirements of the referenced test methods.

## B.3 Variation to barrel construction and support

### B.3.1 General

15. An important aspect of the fire resistance performance of the proposed roller shutter barrel and axle assembly is its ability to support the shutter and its own weight without excessive deformation. This ability will avoid the risk of an opening developing in the slats or, above the shutter curtain that may lead to an integrity failure with respect to the referenced test standards.

### B.3.2 Discussion

**Spanning Capacity of Barrel and Axle**

16. With reference to the tested construction summarised in Appendix A, the barrel assembly performed satisfactorily remained in place and supported the shutter curtain adequately.

17. The proposed construction also incorporates extension of width and height of the roller shutter for periods of exposure 60, 120 and 240 minutes to up to a width of 8m and a height of 8m.

18. The proposed construction incorporates components of various material thickness. However, the difference between the thicknesses are very minor (0.2 mm) and thus it is expected that the temperature of these components will be same as the tested construction. It is therefore reasonable to use the temperatures of the tested construction to predict the temperature of the proposed elements.

19. With reference to BS 5950 Part 8 and the results of the reference tests, the strength retention at 60, 120 and 240 minutes was calculated, and the barrel span section, barrel size, and axle size increased until the stress ratio was close, though not more than, that of the tested construction.

20. For large openings, the spanning capacity of the proposed barrel design was not enough to support the shutter for the full door width. Barrel support frames are proposed to provide additional support for the central section of the barrel. The barrel support frames reduce the effective span of the barrel. As there are two support frames, they will keep the curvature of the barrel to a minimum by not creating a concentrated support at this location.

21. In absence of any introduced strength related weakness of the proposed construction, it is considered the proposed increased widths and barrel section sizes if tested would meet the relevant integrity and impermeability requirements of the referenced test methods for the proposed designs.

**Support Bolts for Barrel and Barrel Support Frames**

22. With reference to the tested construction summarised in Appendix A, the barrel support assembly performed satisfactorily, remained in place and supported the shutter curtain adequately. The shear and tension stress in the bolts were calculated at 240 minutes.

23. The proposed construction also incorporates extension of width and height of the roller shutter for periods of exposure 60, 120 and 240 minutes up to a width of 8m and a height of 8m.

24. This extension increases the load on the supports and the increase in stress which was checked for each design based on the temperature of test components during the test and the stress level at which they were subjected without failure.

25. The loads on the barrel support frames, when present, are the same as for the end supports. Therefore, provided the bolt spacing of the support frames is no less than as for the tested supports, the capacity of the connection will also be the same.

26. In absence of any introduced strength related weakness of the proposed bolted construction it is considered the proposed increased opening width and heights if tested would meet the relevant integrity and impermeability requirements of the referenced test methods for the proposed.